SME206
Signals and Systems

1. Signals and Systems

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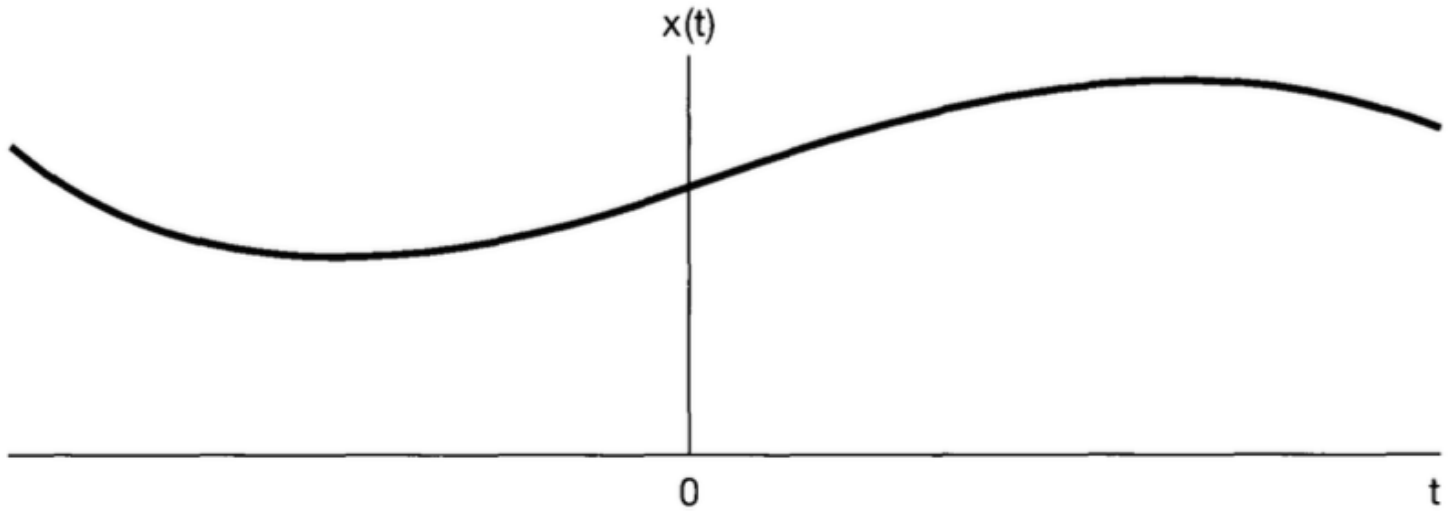
CT & DT Signals: Definition

- **A signal is a quantitative description of a physical phenomenon**, for example,
  - electrical voltage, current in a circuit
  - daily closing value of a share of stock
  - audio signal
- **A signal is a function of one or more variables** (*e.g.* time, space, *etc.*)
  - **In electrical systems, a signal is often a function of one variable in *time***
    - For example, voltage, current, charge, magnetic flux

CT & DT Signals: Mathematical Representation

- In this class, we are interested in two types of signals:

1. **Continuous-time (CT) signal $x(t)$:**
   - where t is a real-valued variable that denotes time: $t \in R$
   - we enclose the independent variable in parenthesis $(\cdot)$

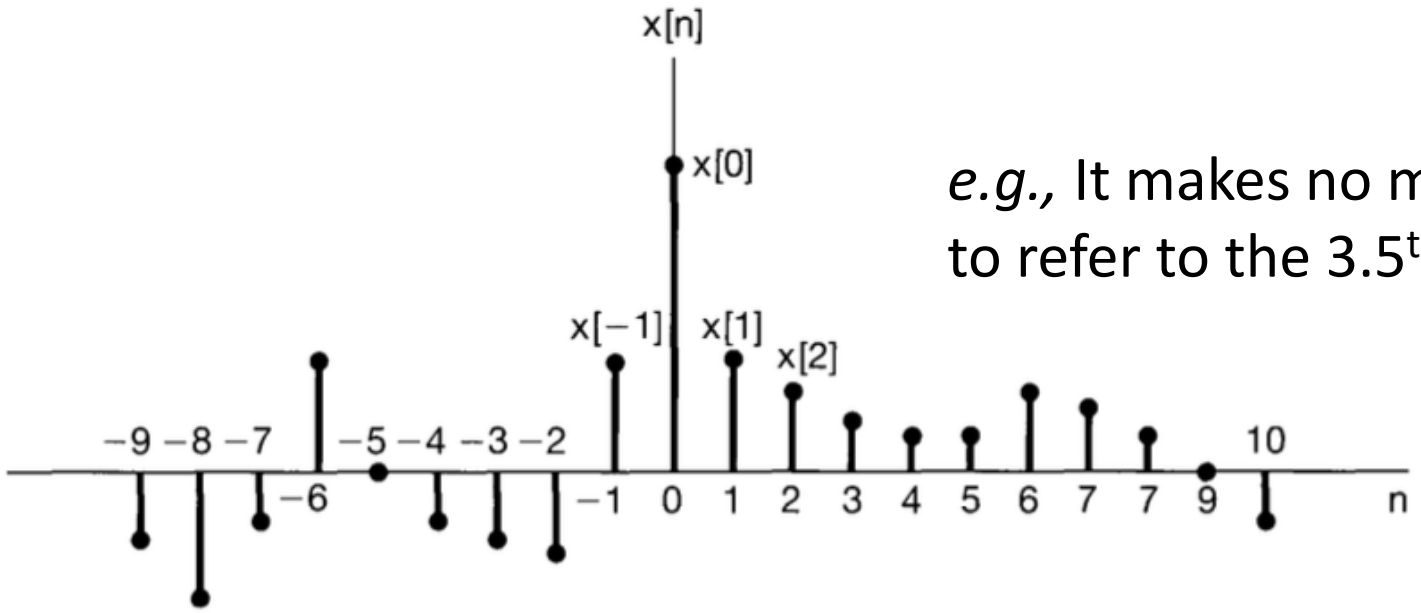


CT & DT Signals: Mathematical Representation

- In this class, we are interested in two types of signals:

2. Discrete-time (DT) signal $x[n]$:

- where n is an integer-valued variable denoting the discrete samples of time: $n \in Z$
- we enclose the independent variable in square brackets $[\cdot]$



e.g., It makes no more sense to refer to the 3.5th sample.

CT & DT Signals: Signal Energy & Power

- In many cases (esp. in this class), the signals we consider are directly related to physical quantities capturing power and energy in a physical system.

- For example, consider a resistor, where $v(t)$ denotes the voltage signal across the resistor, and $i(t)$ denotes the current.
  - Power dissipated by the resistor:

    $$p(t) = v(t)i(t) = \frac{1}{R}v^2(t) = Ri^2(t)$$

    Power is a scaled multiple of the square of the voltage & current signals
  - Energy expended over the time interval $t_1 \leq t \leq t_2$:

    $$\mathrm{E} = \int_{t_1}^{t_2} p(t)dt = \frac{1}{R}\int_{t_1}^{t_2} v^2(t)dt = R\int_{t_1}^{t_2} i^2(t)dt$$
  - Average power dissipated over the time interval $[t_1, t_2]$:

    $$\frac{1}{t_2-t_1}E = \frac{1}{t_2-t_1}\frac{1}{R}\int_{t_1}^{t_2} v^2(t)dt = \frac{R}{t_2-t_1}\int_{t_1}^{t_2} i^2(t)dt$$

CT & DT Signals: Signal Energy & Power

- ***Define*** the energy of an arbitrary signal (potentially complex-valued):
  - CT signal $f(t)$ over a time-interval $[t_1, t_2]$:

    $E_{[t_1,t_2]} \triangleq \int_{t_1}^{t_2} |f(t)|^2 dt$, where $|f(t)|$ denotes the signal magnitude at time t.

  - DT signal $f[n]$ over a time-interval $[n_1, n_2]$:

    $E_{[t_1,t_2]} \triangleq \sum_{n=n_1}^{n_2} |f[n]|^2$, where $|f[n]|$ denotes the signal magnitude at sample n.

- This will end up being a convenient way to measure the "size" of a signal, and may not actually correspond to any physical notion of energy.

CT & DT Signals: Signal Energy & Power

- Often, we are interested in examining the energy of a given signal over an infinite time interval. In this case, we ***define***:
  - CT signal $f(t)$: $E_\infty \triangleq \int_{-\infty}^{\infty} |f(t)|^2 dt$
  - DT signal $f[n]$: $E_\infty \triangleq \sum_{n=-\infty}^{\infty} |f[n]|^2$

- Similarly, for average power over an infinite time interval, we ***define***:
  - CT signal $f(t)$: $P_\infty \triangleq \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |f(t)|^2 dt$
  - DT signal $f[n]$: $P_\infty \triangleq \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |f[n]|^2$

CT & DT Signals: Signal Energy & Power

- Based on the above definitions, we have three classes of signals:
  - Finite energy ($E_\infty < \infty$)
  - Finite average power ($P_\infty < \infty$)
  - Neither finite energy nor average power

* *Note that any signal that has finite energy will also have finite average power (P=0), and that any signal that has a nonzero average power will have infinite energy.*

- A signal with finite energy is said to be an energy signal.
- A signal with (non-zero) finite average power is said to be a power signal.

CT & DT Signals: Signal Energy & Power

Q: classify the following signals:

(1) $f(t) = \begin{cases} e^{-t} \ for \ t \geq 0 \\ 0 \quad for \ t < 0 \end{cases}$;

(2) $f(t) = 1 \ for \ t \in R$;

(3) $f(t) = t \ for \ t \geq 0$.

Transformations of the Independent Variable

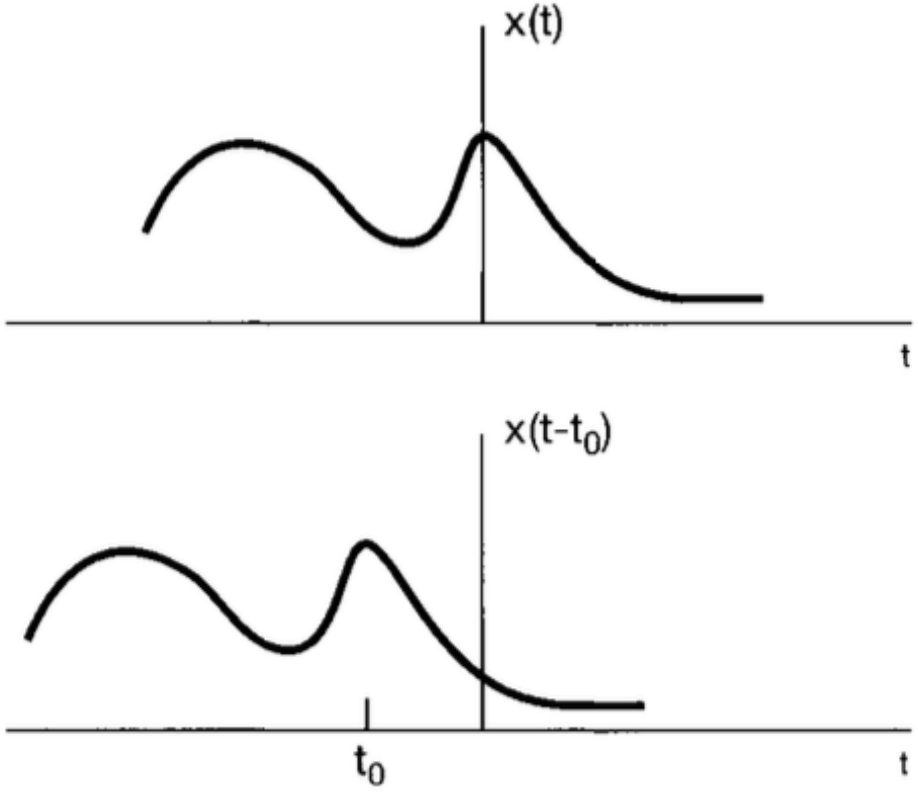
- **<u>Independent variable</u>** (*i.e.,* time) **transformations:**
  1) **Time-shifting**
  2) **Time-reversal**
  3) **Time-scaling**

Transformations of the Independent Variable

- **Independent variable** (*i.e.,* time) **transformations:**

  1) Time-shifting

$$x(t) \rightarrow x(t - t_0)$$



- $t_0 > 0$: signal shifted to right by $|t_0|$ (i.e., delayed in time).
- $t_0 < 0$: signal shifted to left by $|t_0|$ (i.e., advanced in time).

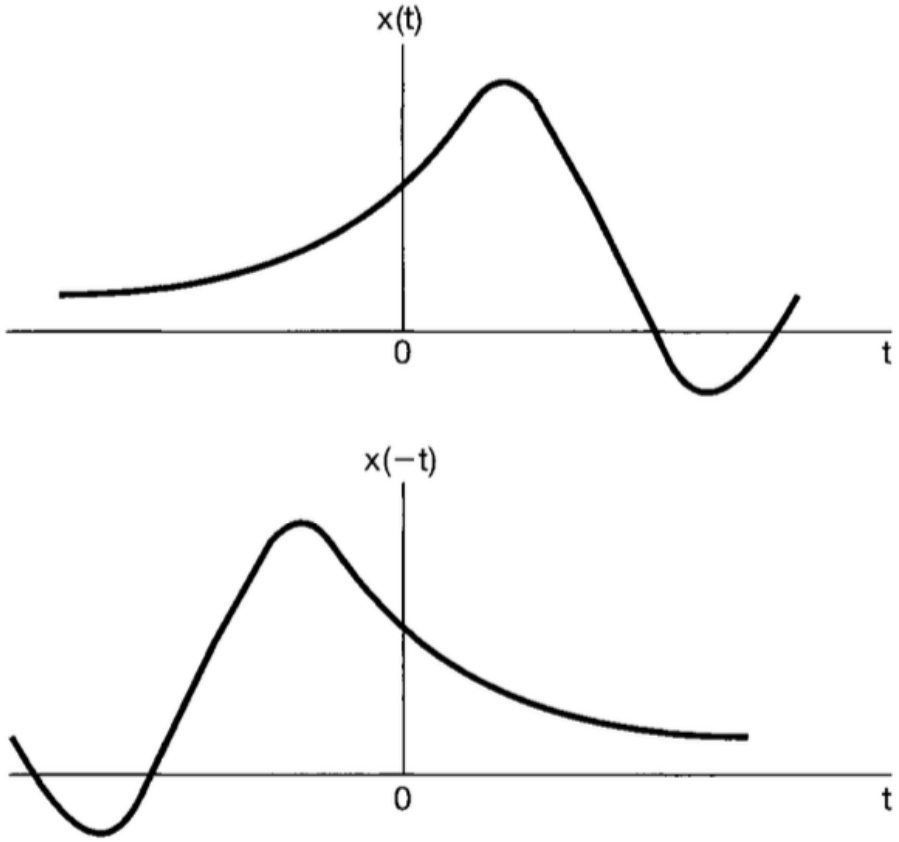
* Analogous definitions hold for DT signal $x[n]$.

Transformations of the Independent Variable

- **<u>Independent variable</u>** (*i.e.,* time) **transformations:**

  2) Time-reversal

$$x(t) \rightarrow x(-t)$$



- Geometrically, $x(-t)$ is a reflection of $x(t)$ about the (vertical) line $t = 0$.

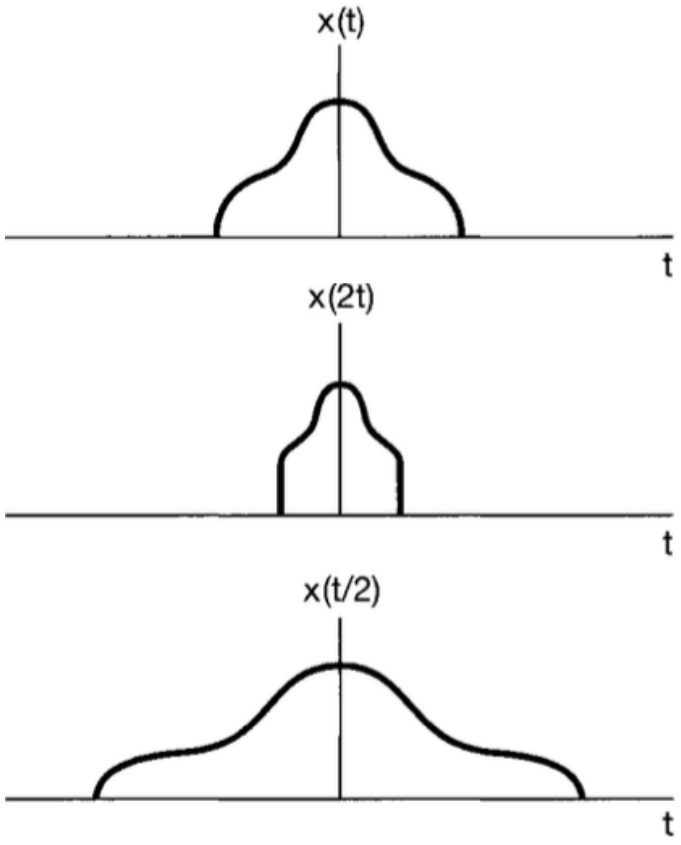
* Analogous definitions hold for DT signal $x[n]$.

Transformations of the Independent Variable

- **Independent variable** (*i.e.,* time) **transformations:**

  3) Time-scaling

$x(t) \rightarrow x(\alpha t)$, where α is a non-zero real number



- If $|\alpha| > 1$, signal compressed along the time axis by a factor of $|\alpha|$.
- If $|\alpha| < 1$, signal expanded/stretched along the time axis by a factor of $\frac{1}{|\alpha|}$.
- If $|\alpha| = 1$, neither compressed nor expanded.
- If $\alpha < 0$, signal is also time reversed.
- *Time-scaling & time-reversal commute.*

* Analogous definitions hold for DT signal $x[n]$.

Transformations of the Independent Variable

- **Independent variable** (*i.e.,* time) **transformations – combined operations**

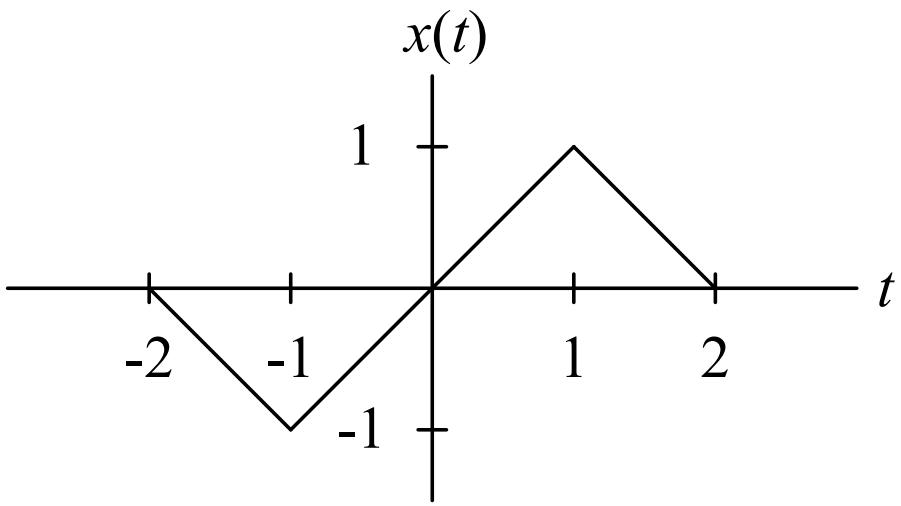
$$g(t) = f(\alpha t + \beta)$$, where α and β are real numbers

  - Since time scaling and time shifting *do not commute*, we must be careful with the order in which these transformations are applied. To draw $g(t)$:
    - First, apply the time-shift β to $f(t)$;
    - Then, apply the scaling α.

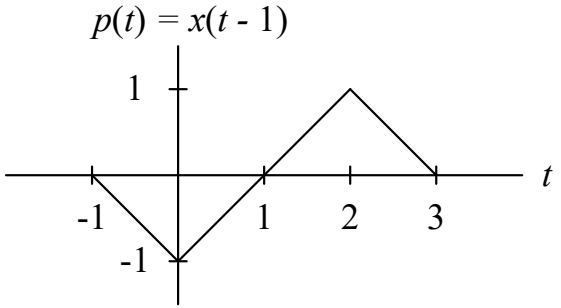
  - Why?

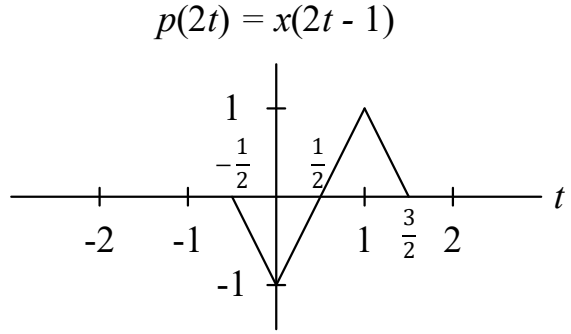
Transformations of the Independent Variable

Q: Given x(t) as shown below, find x(2t - 1)

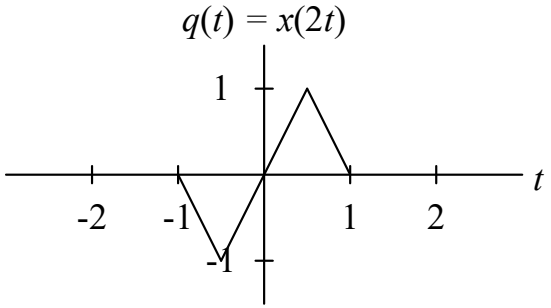


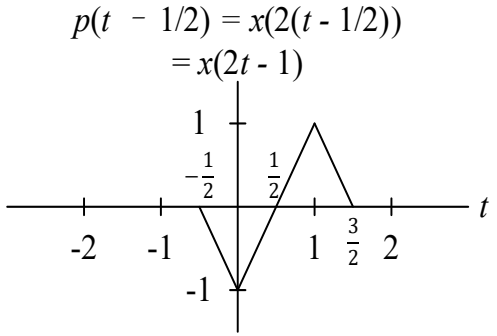
time shift by 1 and then time scale by 2





time scale by 2 and then time shift by $\frac{1}{2}$





Properties of signals: Periodic Signals

- A CT signal $f(t)$ is periodic with period T if $\boldsymbol{f(t) = f(t+T)}$ **for all** $\boldsymbol{t \in R}$.
  - T-Periodic function $\Leftrightarrow$ frequency $\frac{1}{T}$ $\Leftrightarrow$ angular frequency $\frac{2\pi}{T}$

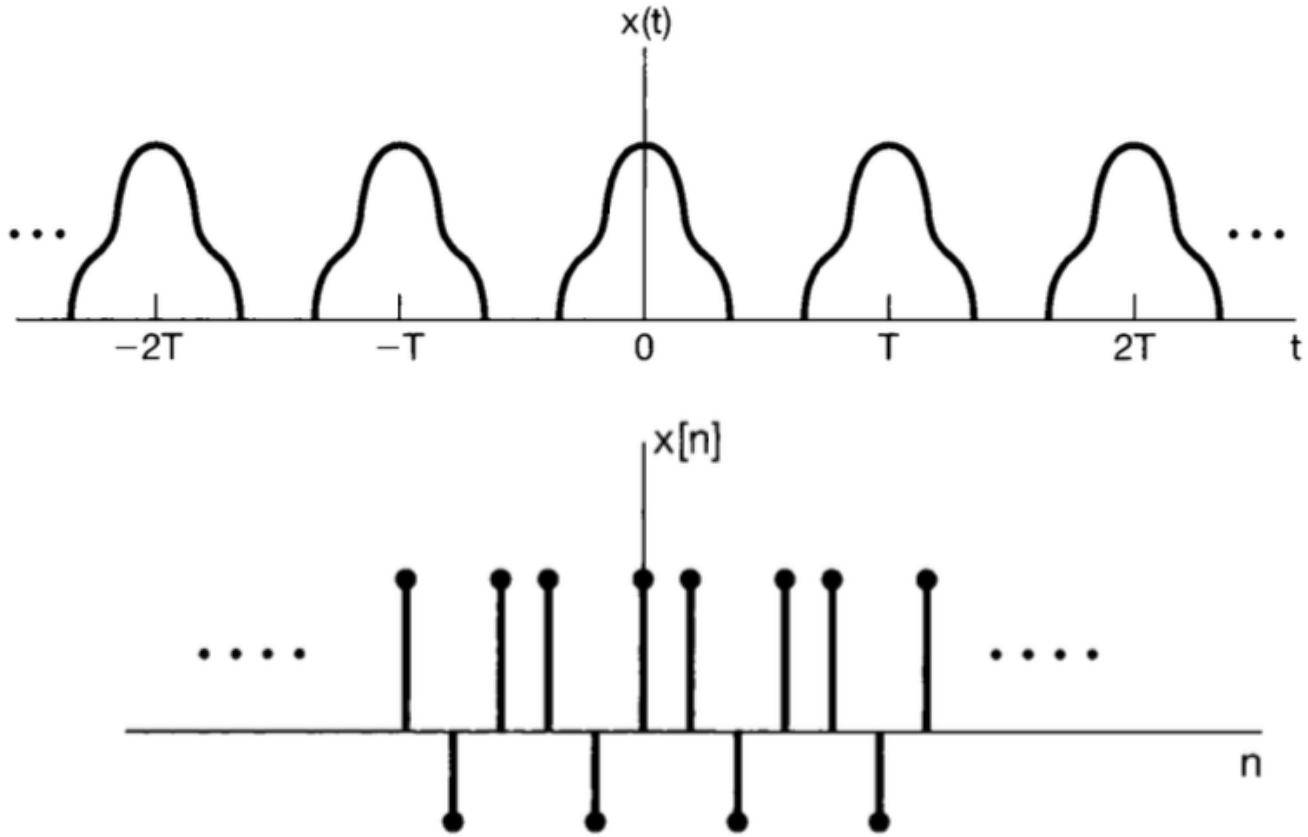
- A DT signal $f[n]$ is periodic with period N if $\boldsymbol{f[n] = f[n+N]}$ **for all** $\boldsymbol{n \in Z}$.
  - N-Periodic sequence $\Leftrightarrow$ frequency $\frac{1}{N}$ $\Leftrightarrow$ angular frequency $\frac{2\pi}{N}$

- **Fundamental period** of a signal: the *smallest period* for which the signal is periodic. The corresponding frequency is called the fundamental frequency.

- A signal that is not periodic is said to be aperiodic.

Properties of signals: Periodic Signals

- Examples



Q: Is the period of a given periodic signal unique?

Q: What is the fundamental period of the above signals?

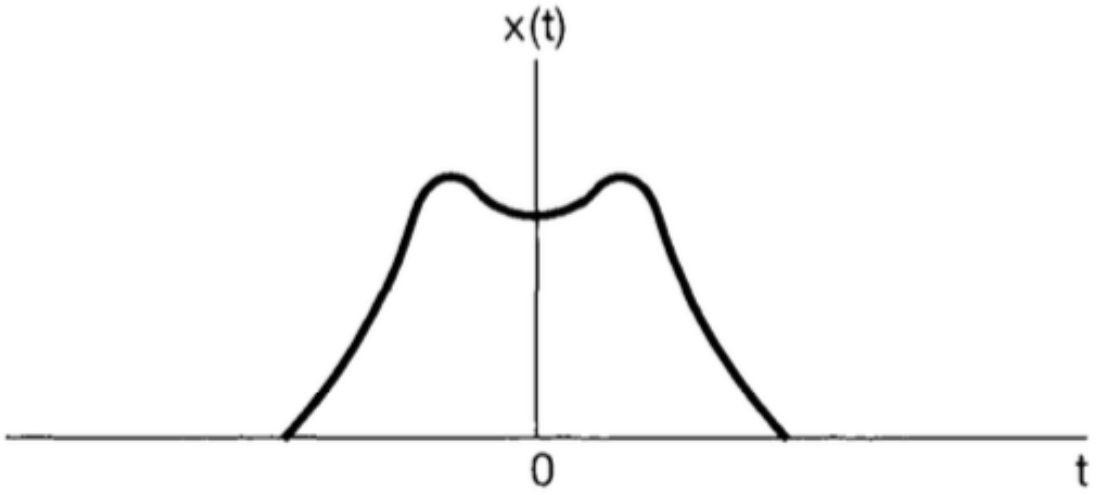
Properties of signals: Even & Odd Signals

- **Even signal:**

$$f(t) = f(-t) \text{ for all } t \in R \text{ (in continuous-time)}$$

$$f[n] = f[-n] \text{ for all } n \in Z \text{ (in discrete-time)}$$

  - Geometrically, an even signal is *symmetric* about the origin.



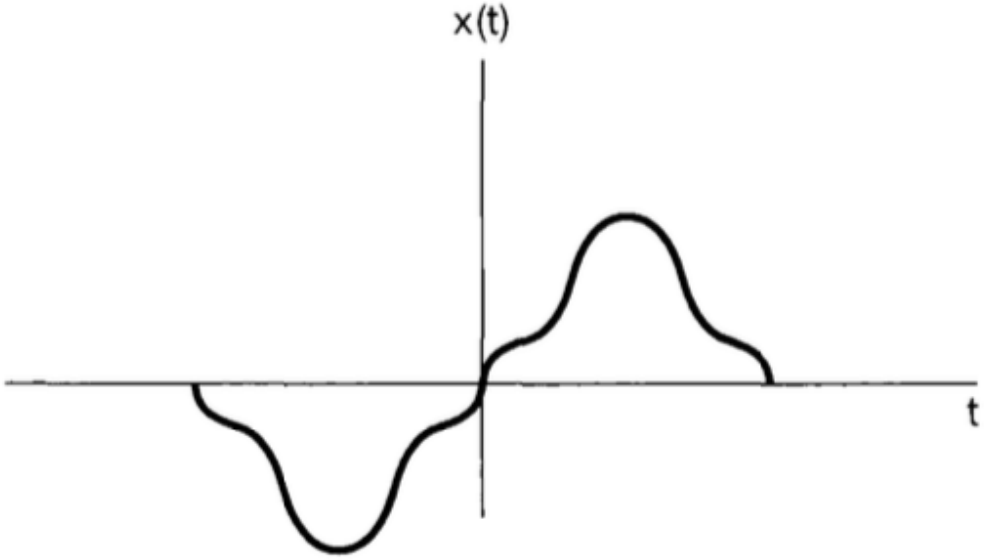
Properties of signals: Even & Odd Signals

- **Odd signal:**

$$\boldsymbol{f(t) = -f(-t)}\ \textbf{for all}\ \boldsymbol{t \in R}\ \textbf{(in continuous-time)}$$

$$\boldsymbol{f[n] = -f[-n]}\ \textbf{for all}\ \boldsymbol{n \in Z}\ \textbf{(in discrete-time)}$$

  - Geometrically, an odd signal is *antisymmetric* about the origin.
  - Note that *an odd signal must necessarily be 0 at* $t = 0$ *or* $n = 0$, since $f(0) = -f(0)$ and $f[0] = -f[0]$.



Properties of signals: Even & Odd Signals

- **Decomposition Theorem:**
  - **Any signal can be broken into a sum of two signals, one of which is even and one of which is odd:**

$$\boldsymbol{x(t) = Ev\{x(t)\} + Od\{x(t)\}}, \text{ where}$$

$$\boldsymbol{Ev\{x(t)\} = \frac{1}{2}[x(t) + x(-t)]}$$

$$\boldsymbol{Od\{x(t)\} = \frac{1}{2}[x(t) - x(-t)]}$$

  - Analogous definitions hold in the discrete-time case.
  - Proof?

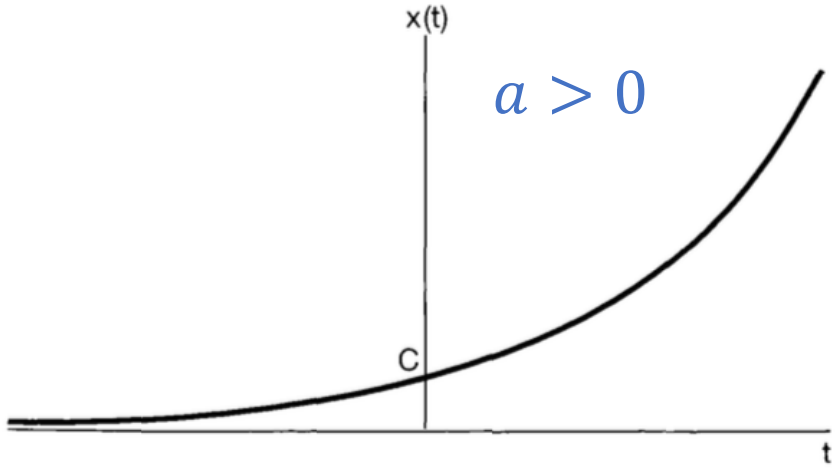
Elementary Signals: CT Complex Exponential Signals

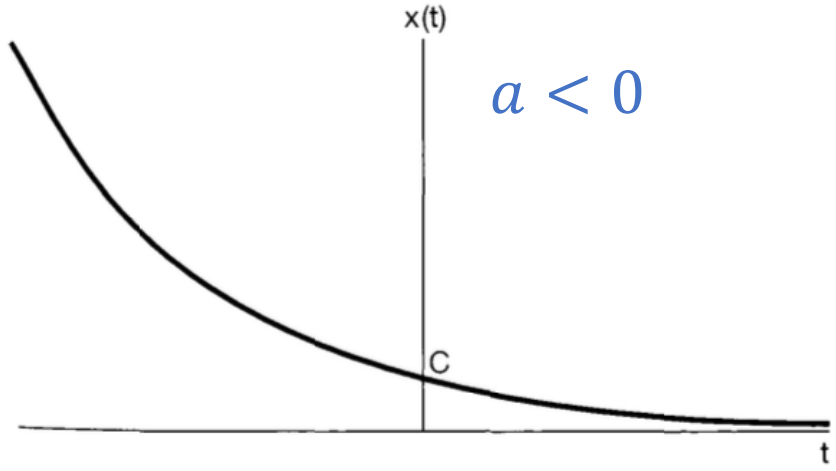
- **The continuous-time *complex exponential signal* is of the form**

  $\boldsymbol{x(t) = Ce^{at}}$**,** where C and a are, in general, complex numbers.

1) Case 1 (real exponential): where $C \in R$ and $a \in R$

- $a > 0$: $x(t)$ is a growing exponential
- $a < 0$: $x(t)$ is a decaying exponential
- $a = 0$: $x(t)$ equals the constant C





Elementary Signals: CT Complex Exponential Signals

- **The continuous-time *complex exponential signal* is of the form**

$$\boldsymbol{x(t) = Ce^{at}}$$, where C and a are, in general, complex numbers.

2) Case 2 (periodic complex exponential): where C is complex and a is purely imaginary

- In other words, $\boldsymbol{x(t) = Ce^{j\omega_0 t}}$, where C is complex and ω_0 is real
- Express C in polar form: $C = |C|e^{j\theta}$, where θ is real
- By Euler's relation, we can re-write $x(t)$ as:

$$\boldsymbol{x(t) = \underbrace{|C|\cos(\omega_0 t + \theta)}_{Re\{x(t)\}} + \underbrace{j|C|\sin(\omega_0 t + \theta)}_{Im\{x(t)\}}}$$

- $x(t)$ is periodic with *fundamental period* $T_0 = \frac{2\pi}{|\omega_0|}$ and *fundamental frequency* $|\omega_0|$. Proof?

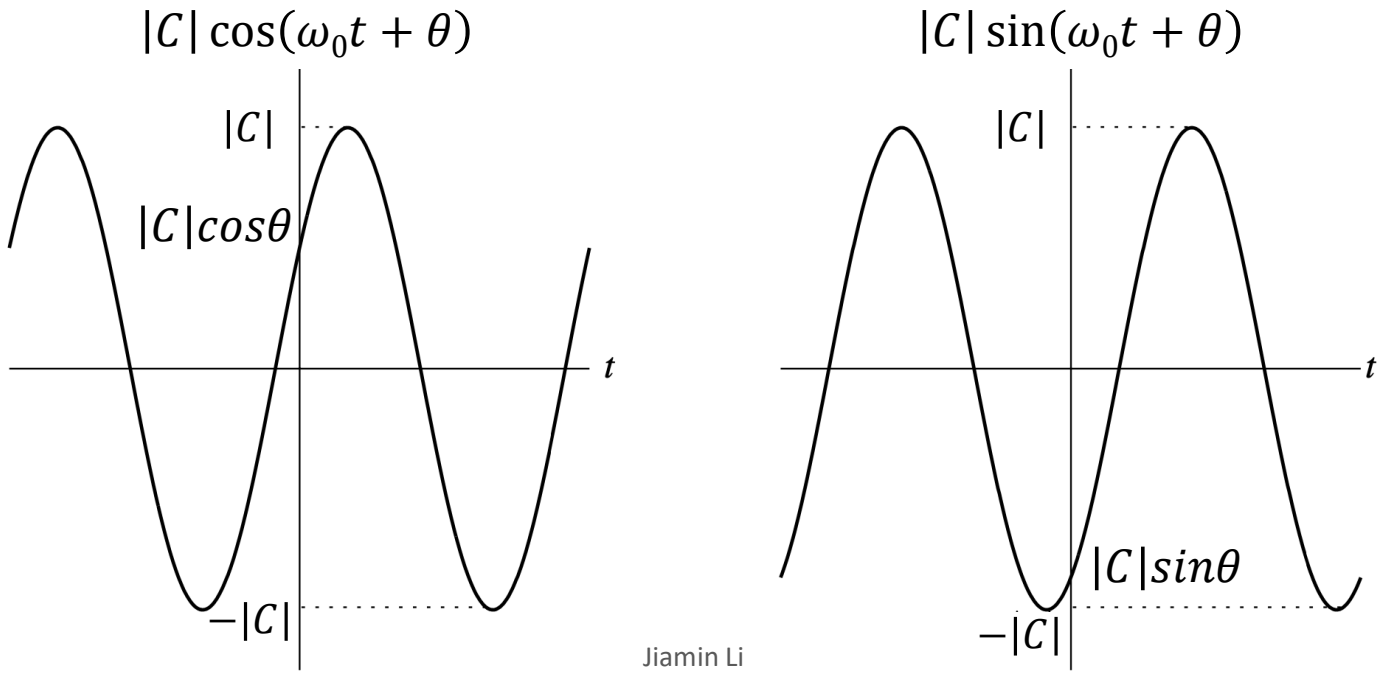
Elementary Signals: CT Complex Exponential Signals

- **The continuous-time *complex exponential signal* is of the form**

$$\boldsymbol{x(t) = Ce^{at}}\text{, where } C \text{ and } a \text{ are, in general, complex numbers.}$$

2) Case 2 (periodic complex exponential): where C is complex and a is purely imaginary

$$\boldsymbol{x(t) = |C|\cos(\omega_0 t + \theta) + j|C|\sin(\omega_0 t + \theta)}$$



Elementary Signals: CT Complex Exponential Signals

2) Case 2 (periodic complex exponential): where C is complex and a is purely imaginary

$$x(t) = |C|\cos(\omega_0 t + \theta) + j|C|\sin(\omega_0 t + \theta)$$

- Given a complex exponential with fundamental frequency ω_0, a *harmonically related set* of complex exponentials is a set of periodic exponentials of the form

$$\boldsymbol{\phi_k(t) = e^{jk\omega_0 t}, k \in Z}$$

- In other words, it is the set of complex exponentials whose frequencies are multiples of the fundamental frequency ω_0.
  - For $k = 0$, $\phi_k(t)$ is a constant;
  - For other values of k, $\phi_k(t)$ is periodic with

    fundamental frequency: $|k\omega_0|$;

    fundamental period: $\frac{2\pi}{|k|\omega_0} = \frac{T_0}{|k|}$.
  - The kth harmonic $\phi_k(t)$ is *still periodic with period* T_0. Proof?

fundamental period $T_0 = \frac{2\pi}{|\omega_0|}$

Elementary Signals: CT Complex Exponential Signals

- **The continuous-time *complex exponential signal* is of the form**

$$\boldsymbol{x(t) = Ce^{at}}$$, where C and a are, in general, complex numbers.

3) Case 3 (general complex exponential): where C and a are complex

- Express C in polar form: $C = |C|e^{j\theta}$, where θ is real
- Express a in rectangular form: $a = r + j\omega_0$, where r, ω_0 are real
- Therefore, we have:

$$x(t) = |C|e^{j\theta}e^{(r+j\omega_0)t} = |C|e^{rt}e^{j(\omega_0 t+\theta)}$$

- By Euler's relation:

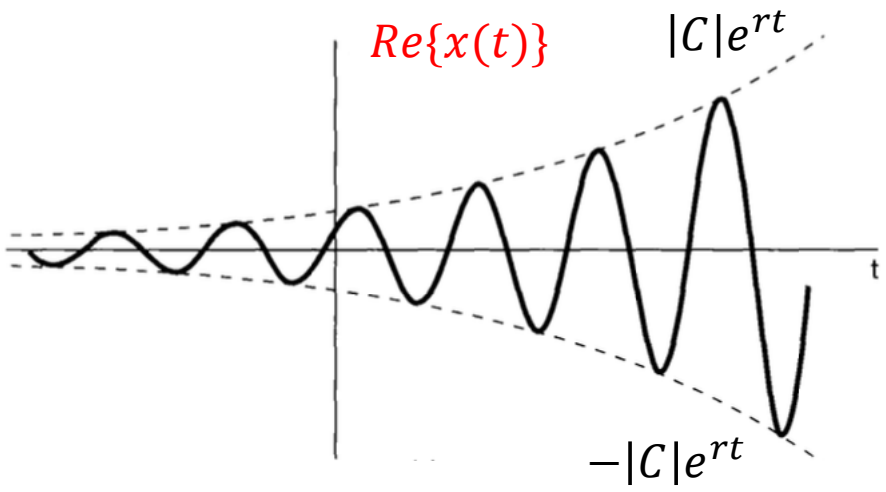
$$x(t) = |C|e^{rt}\cos(\omega_0 t + \theta) + j|C|e^{rt}\sin(\omega_0 t + \theta)$$

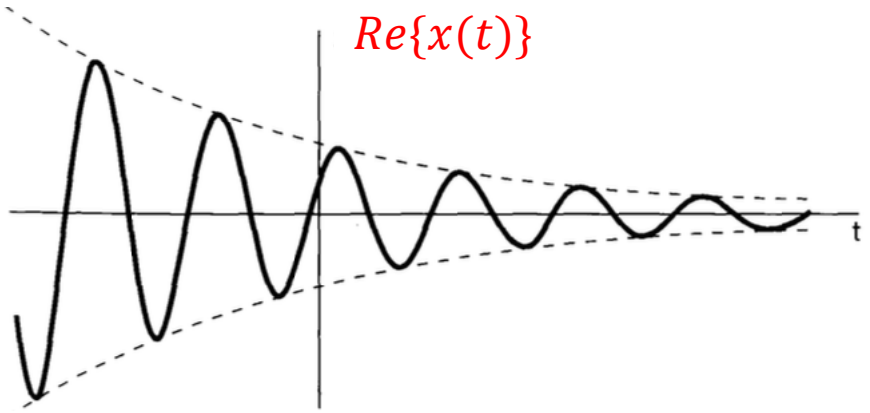
Elementary Signals: CT Complex Exponential Signals

3) Case 3 (general complex exponential): where C and a are complex

$$x(t) = |C|e^{rt}\cos(\omega_0 t + \theta) + j|C|e^{rt}\sin(\omega_0 t + \theta)$$

- $r = 0$ (special case of periodic complex exponential): $Re\{x\}$ and $Im\{x\}$ are sinusoidal.
- $r > 0$: $Re\{x\}$ and $Im\{x\}$ are sinusoidal signals multiplied by a growing exponential.
- $r < 0$: $Re\{x\}$ and $Im\{x\}$ are sinusoidal signals multiplied by a decaying exponential.





Elementary Signals: CT Complex Exponential Signals

- Note on relationship between complex exponentials and sinusoids:
  - From Euler's relation,

$$Ae^{j\omega t} = Acos(\omega t) + jAsin(\omega t)$$

  - Therefore, to express sinusoid by complex exponentials:

$$Acos(\omega t + \theta) = \frac{A}{2}[e^{j(\omega t+\theta)} + e^{-j(\omega t+\theta)}]$$

$$Asin(\omega t + \theta) = \frac{A}{2j}[e^{j(\omega t+\theta)} - e^{-j(\omega t+\theta)}]$$

Elementary Signals: DT Complex Exponential Signals

- **The discrete-time *complex exponential signal* is of the form**

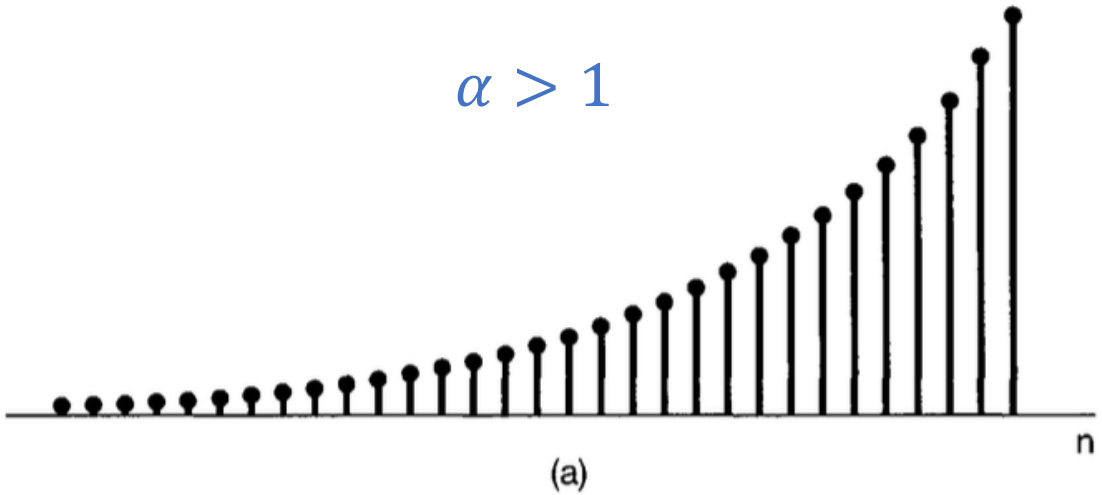
$$x[n] = Ce^{\beta n}, \text{ where } C \text{ and } \beta \text{ are complex numbers.}$$

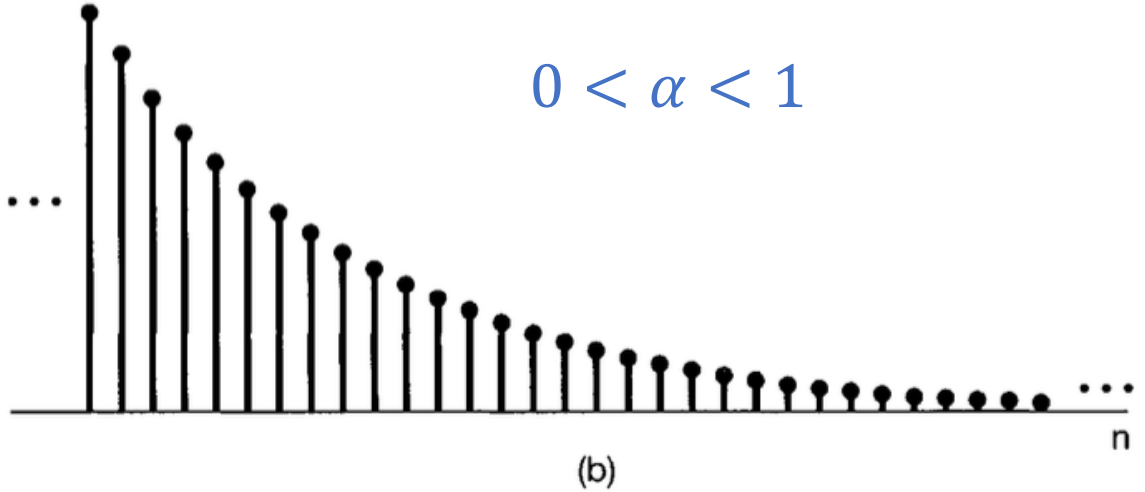
- Let $\alpha = e^{\beta}$, we have $\boldsymbol{x[n] = C\alpha^n}$.

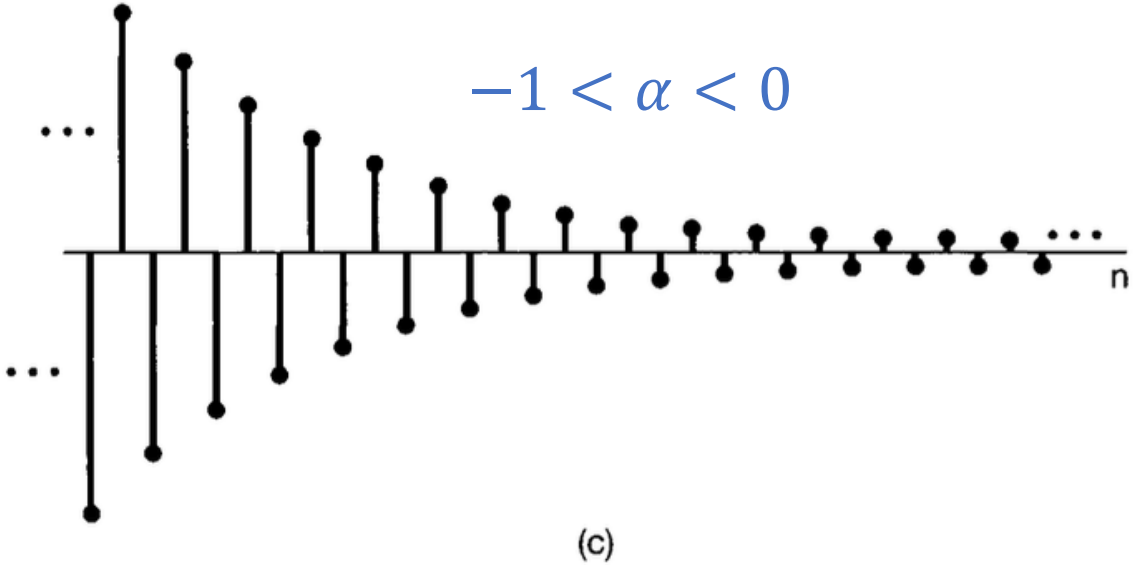
1) Case 1 (real exponential): where $C \in R$ and $\alpha \in R$

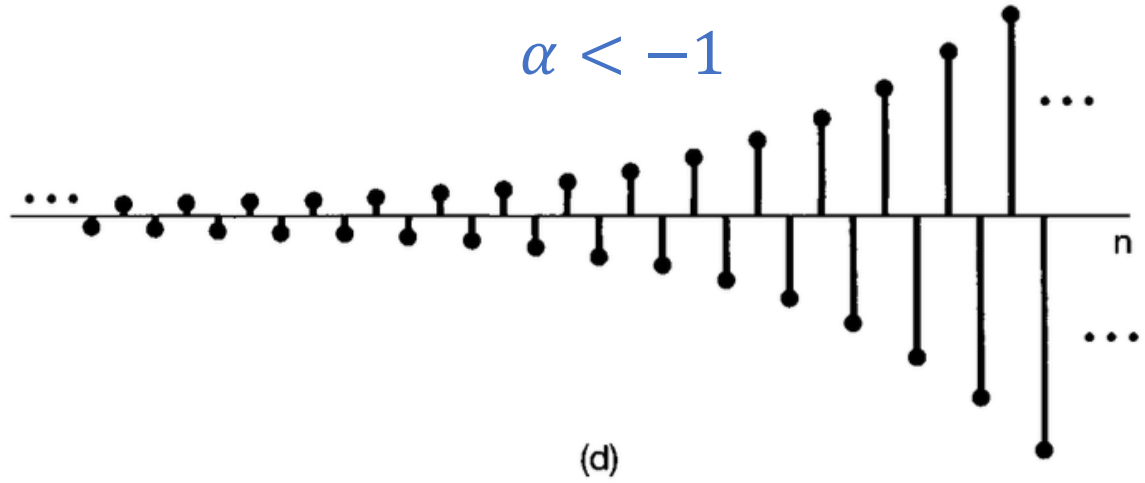
- $|\alpha| > 1$: magnitude of $x[n]$ grows exponentially with n
- $|\alpha| < 1$: magnitude of $x[n]$ decays exponentially with n
- $|\alpha| = 1$: magnitude of $x[n]$ is a constant C
- Note that if $\alpha > 0$, all values of $x[n]$ are of the same sign; if $\alpha < 0$, the sign of $x[n]$ alternates.

Elementary Signals: DT Complex Exponential Signals









Elementary Signals: DT Complex Exponential Signals

- **The discrete-time *complex exponential signal* is of the form**

$$x[n] = Ce^{\beta n}, \text{ where } C \text{ and } \beta \text{ are complex numbers.}$$

- Let $\alpha = e^{\beta}$, we have $\boldsymbol{x[n] = C\alpha^n}$.

2) Case 2 (periodic complex exponential): where β is purely imaginary ($|\alpha| = 1$)

Consider $\boldsymbol{x[n] = Ce^{j\omega_0 n}}$, where $\omega_0 \in R$

- The periodicity condition requires that, for some $N > 0$,

$$x[n+N] = x[n], \text{ where } n \in Z$$

$$\Rightarrow e^{j\omega_0 n} e^{j\omega_0 N} = e^{j\omega_0 n}, \text{ where } n \in Z$$

$$\Rightarrow e^{j\omega_0 N} = 1 \text{ or } \omega_0 N = 2\pi m, \text{ for some } m \in Z$$

Elementary Signals: DT Complex Exponential Signals

- **The discrete-time *complex exponential signal* is of the form**

$$x[n] = Ce^{\beta n}, \text{ where } C \text{ and } \beta \text{ are complex numbers.}$$

- Let $\alpha = e^{\beta}$, we have $\boldsymbol{x[n] = C\alpha^n}$.

2) Case 2 (periodic complex exponential): where β is purely imaginary ($|\alpha| = 1$)

Consider $\boldsymbol{x[n] = Ce^{j\omega_0 n}}$, where $\omega_0 \in R$

- Therefore, the periodicity condition for $x[n]$ is

$$\boldsymbol{\omega_0 = \frac{2\pi m}{N}}, \text{ for some } m \in Z, \text{ and some } N > 0, N \in Z.$$

- Thus $x[n] = e^{j\omega_0 n}$ is periodic if and only if ω_0 is a rational multiple of 2π.
- The fundamental period is $N = \frac{2\pi m}{\omega_0}$, where we assume that m and N are relatively prime, *i.e.*, $\gcd(m, N) = 1$.

Elementary Signals: DT Complex Exponential Signals

Q: Determine the fundamental period of the discrete-time signal:

$$x[n] = e^{j(2\pi/3)n} + e^{j(3\pi/4)n}$$

Q: Determine the fundamental period of the continuous-time signal:

$$x(t) = je^{j10t}$$

Elementary Signals: DT Complex Exponential Signals

- Periodicity property comparison between CT and DT periodic complex exponentials

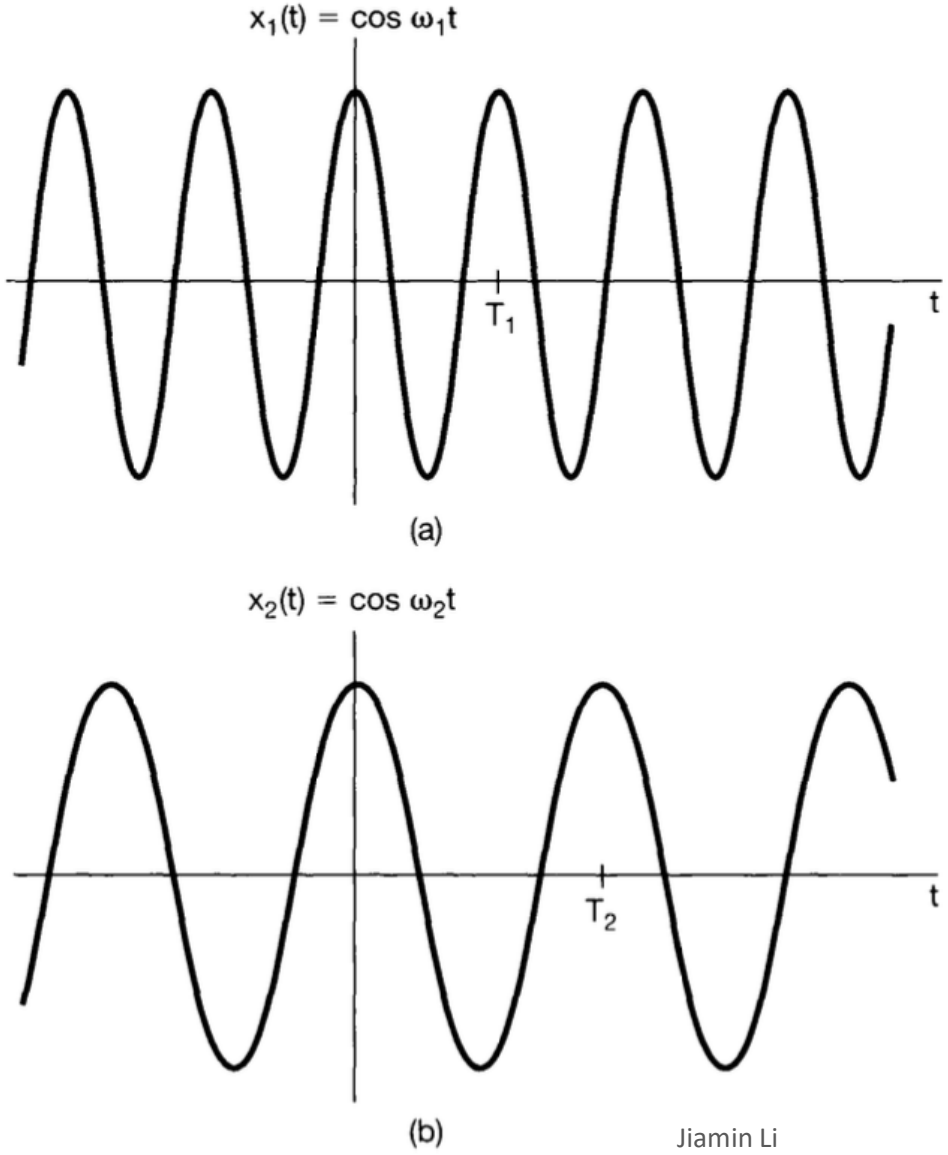
TABLE 1.1 Comparison of the signals $e^{j\omega_0 t}$ and $e^{j\omega_0 n}$.

$e^{j\omega_0 t}$	$e^{j\omega_0 n}$
Distinct signals for distinct values of ω_0	Identical signals for values of ω_0 separated by multiples of 2π
Periodic for any choice of ω_0	Periodic only if $\omega_0 = 2\pi m/N$ for some integers $N > 0$ and m.
Fundamental frequency ω_0	Fundamental frequency* ω_0/m
Fundamental period $\omega_0 = 0$: undefined $\omega_0 \neq 0$: $\frac{2\pi}{\omega_0}$	Fundamental period* $\omega_0 = 0$: undefined $\omega_0 \neq 0$: $m\left(\frac{2\pi}{\omega_0}\right)$

*Assumes that m and N do not have any factors in common.

Why?

Elementary Signals: DT Complex Exponential Signals



$$\omega_1 > \omega_2$$
$$T_1 < T_2$$

Elementary Signals: DT Complex Exponential Signals

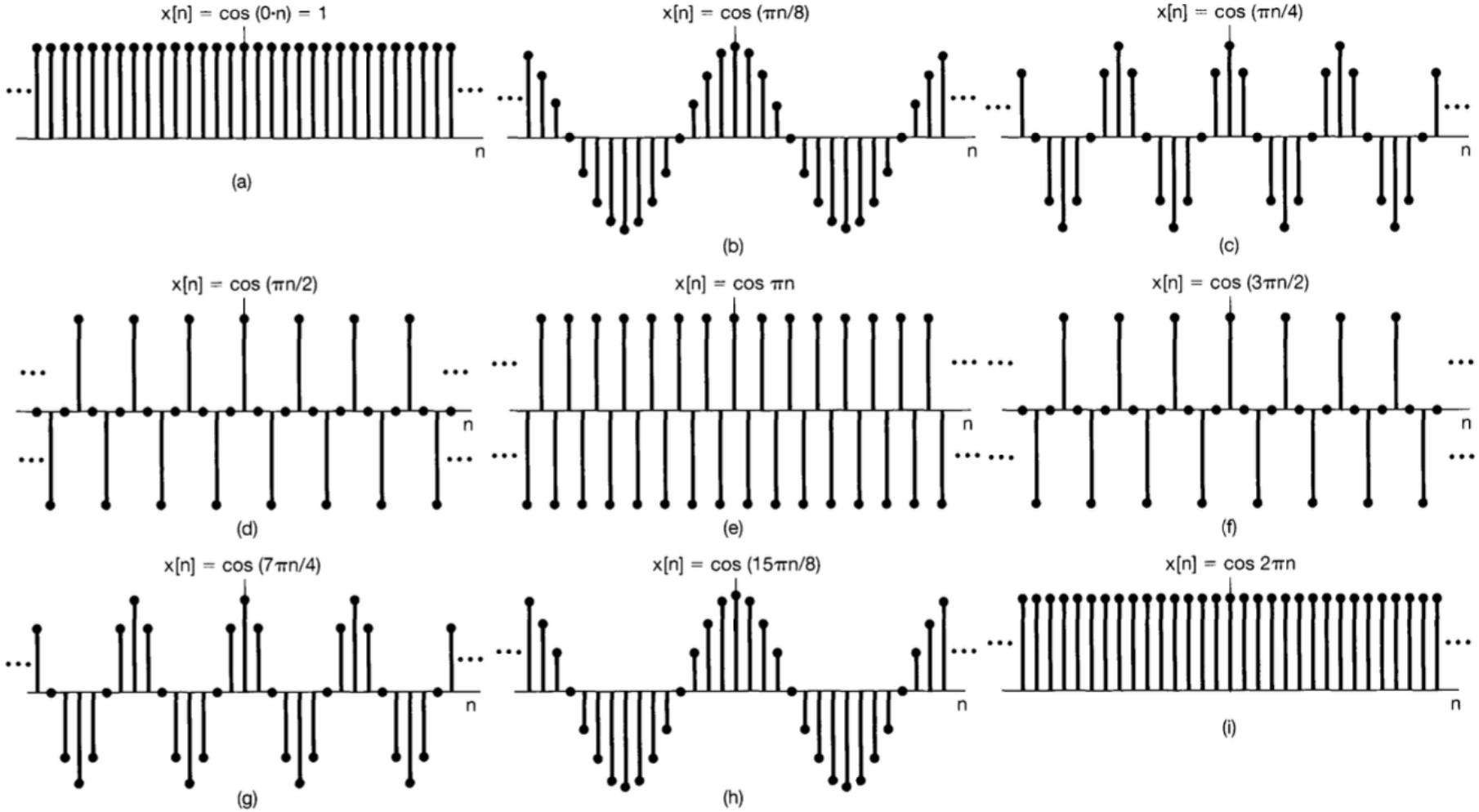


Figure 1.27 Discrete-time sinusoidal sequences for several different frequencies.

As ω_0 increases in magnitude, the signal $e^{j\omega_0 n}$ does not have a continually increasing rate of oscillation!

Elementary Signals: DT Complex Exponential Signals

2) Case 2 (periodic complex exponential): where β is purely imaginary ($|\alpha| = 1$)

Consider $\boldsymbol{x[n] = Ce^{j\omega_0 n}}$, where $\omega_0 \in R$

- Therefore, the periodicity condition for $x[n]$ is

$$\boldsymbol{\omega_0 = \frac{2\pi m}{N}}, \text{ for some } m \in Z, \text{ and some } N > 0, N \in Z.$$

- Sets of harmonically related periodic exponentials: the signals of frequencies which are multiples of $2\pi/N$. That is,

$$\boldsymbol{\phi_k[n] = e^{jk\left(\frac{2\pi}{N}\right)n}, k \in Z}$$

- *Are all of the harmonically related DT complex exponentials distinct?*
  - There are only $\boldsymbol{N}$ distinct periodic exponentials in the set. For example,

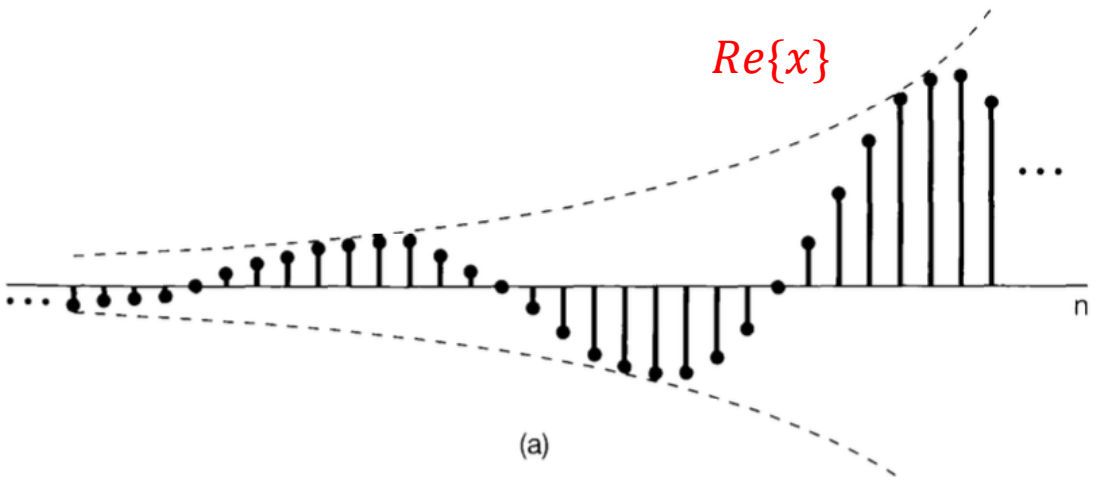
$$\phi_0[n] = 1, \phi_1[n] = e^{\frac{j2\pi n}{N}}, \phi_{2[n]} = e^{\frac{j4\pi n}{N}}, \phi_{N-1}[n] = e^{j2\pi(N-1)n/N}$$

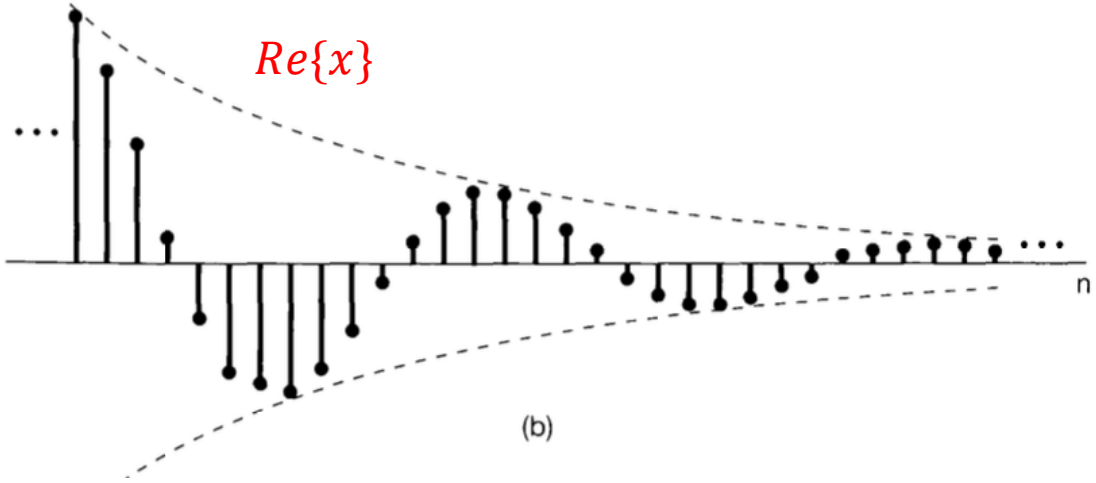
  - Any other $\phi_k[n]$ is identical to one of these.

Elementary Signals: DT Complex Exponential Signals

3) Case 3 (general complex exponential): where C and α are complex

- We write: $C = |\mathrm{C}|e^{j\theta}$, and $\alpha = |\alpha|e^{j\omega_0}$,
- Then, $x[n] = |C||\alpha|^n \cos(\omega_0 n + \theta) + j|C||\alpha|^n \sin(\omega_0 n + \theta)$
  - $|\alpha| = 1$: then $x[n] = |C| \cos(\omega_0 n + \theta) + j|C| \sin(\omega_0 n + \theta)$. It has sinusoidal real and imaginary parts (*not necessarily periodic, though*)
  - $|\alpha| > 1$: then $Re\{x\}$ and $Im\{x\}$ are multiplied by a growing exponential
  - $|\alpha| < 1$: then $Re\{x\}$ and $Im\{x\}$ are multiplied by a decaying exponential

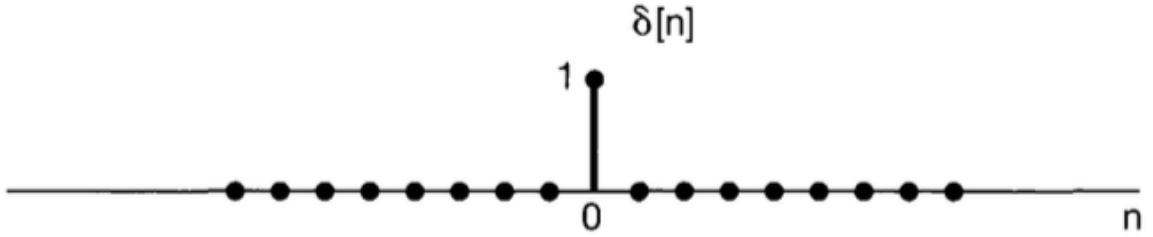




Unit Impulse & Unit Step Functions: DT

- The *discrete-time unit impulse signal* $\delta[n]$ is **defined** as:

$$\delta[n] = \begin{cases} 1, & n = 0, \\ 0, & n \neq 0. \end{cases}$$



- The *discrete-time unit step signal* $u[n]$ is **defined** as:

$$u[n] = \begin{cases} 1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$



- It can be shown that:

$$\delta[n] = u[n] - u[n-1]$$

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$

$$u[n] = \sum_{k=-\infty}^{\infty} u[k]\delta[n-k]$$

Unit Impulse & Unit Step Functions: DT

- **Sampling Property (or Sifting Property):**
  - By the definition of $\delta[n]$, $\delta[n-n_0]=1$ if $n=n_0$, and 0 otherwise. Thus,

$$x[n]\delta[n-n_0]=\begin{cases}x[n], & n=n_0\\ 0, & n\neq n_0\end{cases}$$
$$=x[n_0]\delta[n-n_0]$$

    - In other words, when a signal $x[n]$ is multiplied with $\delta[n-n_0]$, the output is a unit impulse with amplitude $x[n_0]$.
- **Representation Property:**

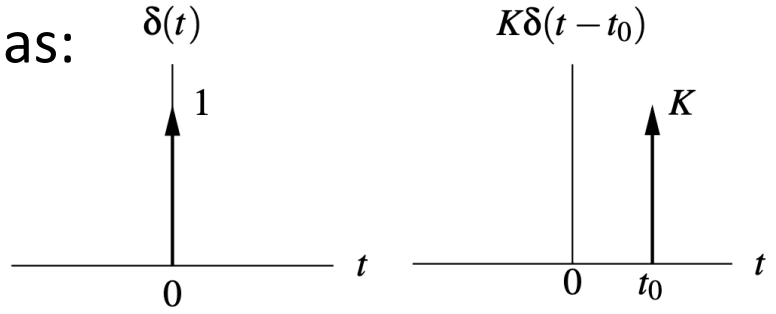
$$x[n]=\sum_{n_0=-\infty}^{\infty}x[n_0]\,\delta[n-n_0]$$

  - In other words, any signal $x[n]$ can be written as a sum of scaled and shifted impulse signals.

Unit Impulse & Unit Step Functions: CT

- The *continuous-time unit impulse function* $\delta(t)$ is **defined** as:

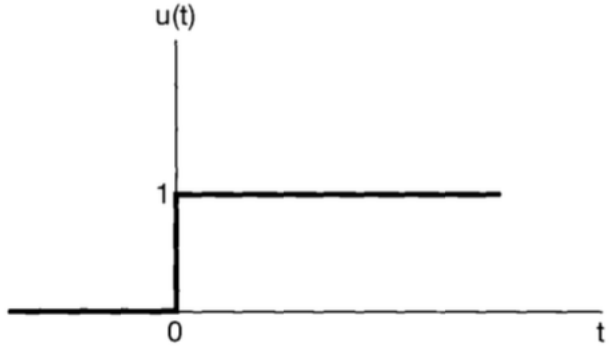
$$\delta(t) = \begin{cases} 0, & t \neq 0, \\ \infty, & t = 0. \end{cases}, \quad \text{where } \int_{-\infty}^{\infty} \delta(t)dt = 1.$$



- The *continuous-time unit step function* $u(t)$ is **defined** as:

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases}$$

  - Note that $u(t)$ is discontinuous at $t = 0$.



- Relations between $\delta(t)$ and $u(t)$:

$$\delta(t) = \frac{du(t)}{dt}$$

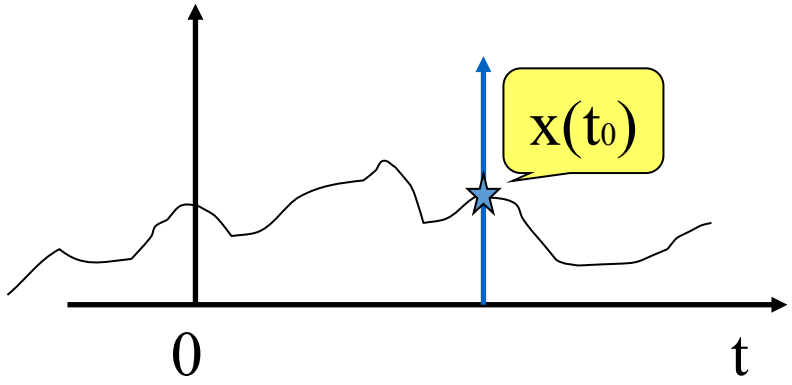
$$u(t) = \int_{-\infty}^{t} \delta(\tau)d\tau$$

Unit Impulse & Unit Step Functions: CT

- **Sampling property (sifting property):**

$$x(t)\delta(t) = x(0)\delta(t)$$

$$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$$



- Following the sampling property, we have:

$$\int_{-\infty}^{\infty} x(t)\delta(t)dt = \int_{-\infty}^{\infty} x(0)\delta(t)dt = x(0)\int_{-\infty}^{\infty} \delta(t)dt = x(0)$$

$$\int_{-\infty}^{\infty} x(t)\delta(t - t_0)dt = x(t_0)$$

Unit Impulse & Unit Step Functions: CT

- **Representation Property:**

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$$

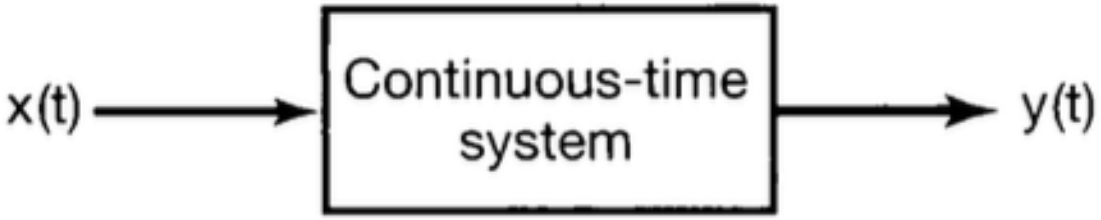
  - As a special case:

$$u(t) = \int_{-\infty}^{\infty} u(\tau)\delta(t-\tau)d\tau = \int_{0}^{\infty} \delta(t-\tau)d\tau,$$

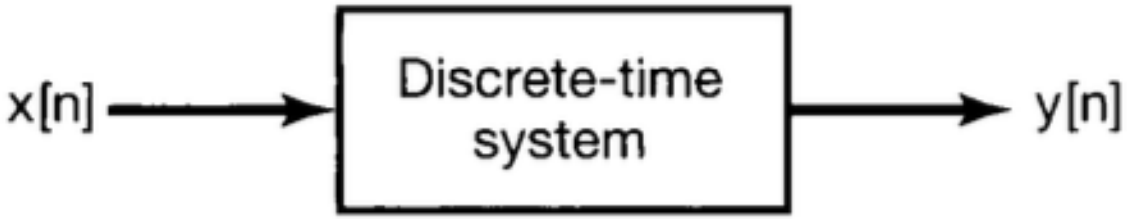
$$\Rightarrow u(t) = \int_{-\infty}^{t} \delta(\sigma)d\sigma, \text{ let } \sigma = t-\tau$$

CT & DT Systems: Definition

- A **system** is a quantitative description of a physical process that *transforms signals (at its "input") to signals (at its "output")*.



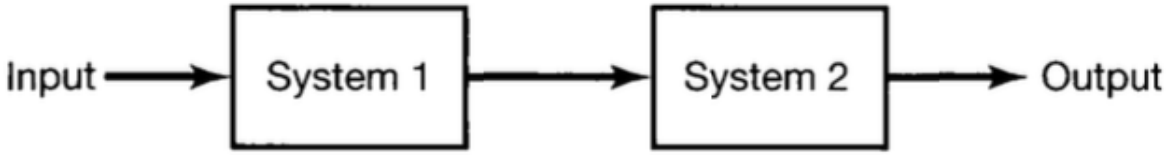
Input-output relation of a CT system: $x(t) \longrightarrow y(t)$



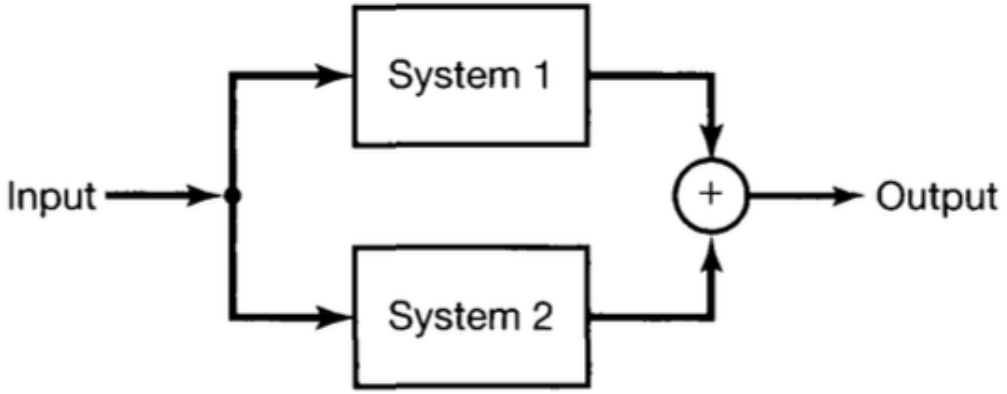
Input-output relation of a DT system: $x[n] \longrightarrow y[n]$

CT & DT Systems: Interconnections

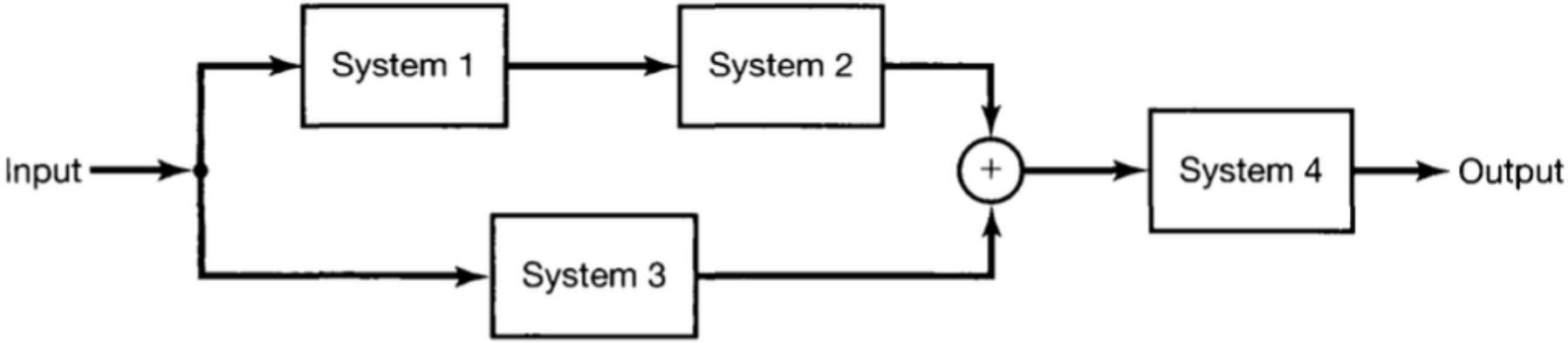
- **Basic ways of system interconnections**



Series (cascade) interconnection



Parallel interconnection



Series-parallel interconnection

CT & DT Systems: Properties

- **Memory**
  - A system is *memoryless* if its output at time t (or n) depends only on the input at that same time (at time t, or n).
  - Otherwise, it is said to have memory.

Q: Identify the memory property of the following system:

1) $y(t) = (x(t) + x^2(t))^2$
2) $y[n] = x[n-1]$
3) $y[n] = x[n] - y[n-1]$

CT & DT Systems: Properties

- **Invertibility**
  - A system is *invertible* if distinct inputs lead to distinct outputs.
    - In other words, there exists a one-to-one mapping from the set of input signals to the set of output signals.



  - To show the invertibility, one has to show the formula.
  - To show otherwise, a counter example will do.

Q: Identify the invertibility property of the following system:

1) $y(t) = (\cos(t) + 2)x(t)$

2) $y[n] = \sum_{k=-\infty}^{n} x[k]$

3) $y(t) = x^2(t)$

CT & DT Systems: Properties

- **Causality**
  - A system is *causal* if the output at any time depends on values of the input at only the present and past times. (also known as being *non-anticipative*)

Q: Identify the causality property of the following system:

1) $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$
2) $y[n] = x[n] + x[n+1]$
3) $y[n] = \frac{1}{2M+1}\sum_{k=-M}^{+M} x[n-k]$

CT & DT Systems: Properties

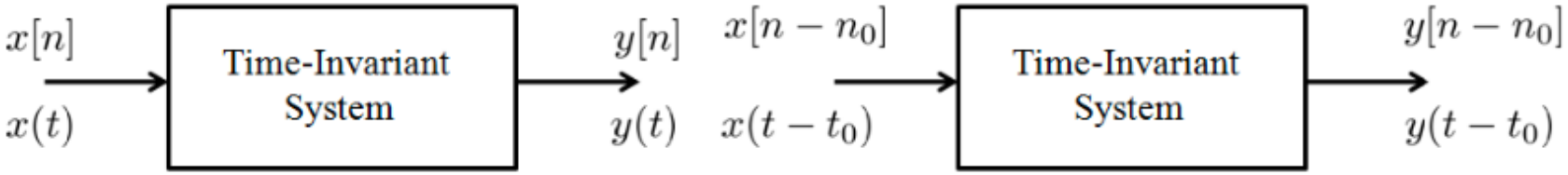
- **Stability**
  - A system is *stable* if a *bounded input* always produces a *bounded output signal*. That is, if $|x(t)| \leq B$ for some $B < \infty$, then $|y(t)| < \infty$.
  - A signal $x(t)$ (and $x[n]$) is bounded if there exists a constant $B < \infty$ such that $|x(t)| \leq B$ for all t.

Q: Identify the stability property of the following system:

1) $y(t) = 2x^2(t-1) + x(3t)$
2) $y[n] = \sum_{k=-\infty}^{n} u[k]$

CT & DT Systems: Properties

- **Time Invariance**
  - A system is *time-invariant* if a time-shift of the input signal results in the same time-shift of the output signal.
    - In other words, if $x(t) \longrightarrow y(t)$, then $x(t - t_0) \longrightarrow y(t - t_0)$.



Q: Identify if the following system is time-invariant:

1) $y(t) = \sin(x(t))$
2) $y[n] = nx[n]$

CT & DT Systems: Properties

- **Linear**
  - A system is *linear* if it is additive and scalable.
    - Additive: $x_1(t) + x_2(t) \longrightarrow y_1(t) + y_2(t)$
    - Scalable: a$x_1(t) \longrightarrow ay_1(t)$, where a is any complex constant
    - The above two properties can be *combined into a single statement*:

$$ax_1(t) + bx_2(t) \longrightarrow ay_1(t) + by_2(t), \text{ for all } a, \text{b} \in C.$$

Q: Identify if the following system is linear:

1) $y(t) = x^2(t)$
2) $y[n] = Re\{x[n]\}$

Summary

- **Definitions of Signals**
  - Continuous-time & discrete-time signals, mathematical representation
  - Signal energy & power
  - Transformations of the independent variable: time-shifting, time-reversal, time-scaling, combined operations
  - Periodic, even, odd signal
- **Elementary Signals**
  - CT & DT complex exponential signal
  - Periodic property, fundamental frequency
  - Unit impulse & unit step functions, the sampling & representative property
- **Definition of Systems**
  - Continuous-time & discrete-time systems, system interconnections
  - System properties: memoryless, invertibility, causality, stability, time invariance & linearity